Data-driven regional traffic simulation from heterogeneous mobility sources

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Abstract

Regional traffic simulators increasingly draw on mobile-phone demand, loop-detector counts and GPS traces, but they are rarely validated on independent data, and the contribution of each source is seldom measured. We present an agent-based simulator of ordinary traffic on the regional road network of Emilia-Romagna, Italy, in which each source has one role: mobile-phone origin–destination tables give the demand and its timing, half of the loop detectors set the demand scale, and GPS traces from one province set route weights by road class. We evaluate it on held-out detectors in winter 2020 and summer 2021, on GPS link counts and in a leave-one-detector-out test of a local correction, always against predictors without simulation, and we measure the change brought by each source. On held-out detectors the simulated total volume is within 1% of the observed one in winter and 10% in summer, and roads are ranked better than by a road-class predictor (Spearman correlation 0.67–0.68 against 0.41–0.48). The daily timing comes mostly from the demand profile and does not improve on the mean detector profile. GPS-informed route choice improves the ranking of roads, province factors reduce the regional bias, and the local correction lowers the hourly error at unseen detectors by 2.4 and 6.0 percentage points. The error at individual roads remains comparable to that of simple predictors, and the value of specific origin–destination pairings could not be determined with the available statistical power. Aggregate level, spatial ranking, timing, unmonitored roads and source contributions are treated as distinct evaluation objectives.

Keywords: Agent-based simulation, Simulation validation, Traffic reconstruction, Mobile phone data, Loop detector, GPS probe data

1. Introduction

Transport planning [1], the study of congestion [2] and the assessment of emissions [3] all need to know how much traffic each road carries, hour by hour. No single data source provides this. Mobile phones show where people travel but not which roads they use, and phone samples have their own biases [4, 5, 6]. Loop detectors count every vehicle, but only

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at a few points, and where they are placed limits what we can learn about the rest of the network [7, 8, 9, 10]. GPS-equipped vehicles cover many roads, but they are a small sample of traffic, and the chance of sampling a vehicle changes between roads and hours [11, 12, 13]. A regional model therefore has to combine these sources [14, 15, 16, 17]. This raises a simple question: what does each of them actually add?

Each source has been studied for a long time. Phone and GPS data have shown statistical regularities in how individuals move [18, 19, 20, 21, 22, 23]. Phone records have also been used to estimate origin–destination (OD) matrices [24, 25], and these matrices have been assigned to road networks [26] to study road use and congestion [27, 1, 2]. Detector counts have long been used to estimate OD matrices [28, 29, 30] and to calibrate agent-based simulations [31]. GPS probes and fixed sensors are used to estimate the state of traffic, mostly on motorways and urban corridors [32, 33, 34, 35, 36, 37].

More recent work combines the sources. Telecom data and vehicle flows have been used together to predict traffic where there is no camera [38], traffic has been estimated at places without sensors [39], and a few fixed sensors have been scaled up to describe whole urban networks [40]. Closer to our work, agent-based simulations now take their demand from phone data. In Barcelona, activity diaries built from phone records feed a MATSim model [41]. In Novi Sad, telecom data are combined with population and GIS data, and traffic is calibrated and validated with vehicle counters [3]. Aggregated phone data have also been used to build a model of Brussels [42]. A recent review of more than 60 such case studies shows how common these data-driven, city-wide simulations have become [43]. Combining the sources is therefore no longer new, but doing so does not by itself show what each of them adds.

Three things make this hard to see. First, independent data are scarce. The simulation literature separates verification, whether a model is implemented as intended, from validation, whether it is accurate enough for its purpose, and asks that validation use data not used to build the model [44, 45]; the same holds for traffic simulation [46, 47]. Yet demand is often rescaled to match the counts, and agreement at those same counts is then reported. The closest studies show how few options there can be: Çolak et al. report that no single validation source existed for their cities [2], Toole et al. compare their estimates with travel surveys [1], and Bassolas et al. keep traffic counts out of the calibration but have them only

at some main entrances of the city [41]. Second, the sources are usually tested together. The result then shows whether the complete model works, not which source made it work, even though uncertainty in demand affects every stage of a model [48, 49]. Third, a model needs a benchmark. Simple historical profiles predict the daily pattern of traffic very well [50, 51], so good timing means little unless a model does better than they do.

In this paper we build and test such a simulator for a whole region, Emilia-Romagna, Italy, with hourly traffic on the regional road network. We give each source one job in the model and check that job with data that played no part in it (Table 1). The contributions are methodological as much as regional:

- an assessment protocol that keeps apart five objectives that are often reported together: the aggregate level of traffic, the ranking of roads, the daily timing, the estimate at roads without detectors, and the contribution of each source;
- a measure of the marginal information of each source, the change on held-out data when that source is added, changed or removed, with the parameters that act through it held fixed;
- a comparison of every output with predictors that use no simulation, so that agreement with the data is not mistaken for skill;
- an account of what the evidence does not establish, including the error at single roads and the value of specific origin–destination pairings.

Half of the detectors scale the demand, with one factor per day type and one per province. Route choice uses GPS measured in one province and transferred to the rest of the region by road class. We compare timing on the GPS-matched roads of Forlì–Cesena, volumes at held-out detectors in two seasons, and a local correction by leaving out each detector in turn, always against predictors that use no simulation. Section 2 describes the data and Section 3 the model and the tests. Section 4 follows the model as it is built: first the level of traffic, then its timing, the ranking of roads and the correction, then the error at single roads, and finally what each source adds. Section 5 discusses what this means.

Figure 1 shows how the data enter the simulator and which data are used to assess each output.

Table 1: Role of each data source. Each row gives one data set, the part of the model it sets and the comparison it serves. Each step is compared with data that did not enter it; the depth of the correction is the one exception, explained in note *a*.

Data	Enters the model in	Used for comparison in
Mobile-phone OD (Jan–Feb 2020)	where and when trips occur	–
GPS traces, Forlì–Cesena	route choice (class speeds and usage)	–
GPS link counts, Forlì–Cesena, two campaigns	–	hourly profiles on the matched links
Detectors, half of February 2020	demand scale k, province factors, trip radius, departure shift, cost weights, depth[a]	–
Detectors, other half of February 2020	local correction; adoption of province factors[b]	volume level and ranking (2020)
Detectors, June–July 2021	local correction; adoption of province factors[b]	volume level and ranking, later season
Detector left out of the correction	–	leave-one-detector-out test

[a]The depth of the correction was selected by cross-validation over all detectors of 2020; 2021 played no part in that selection. [b]These detectors decided once whether the factors, set on the calibration half, were kept; they set no value.

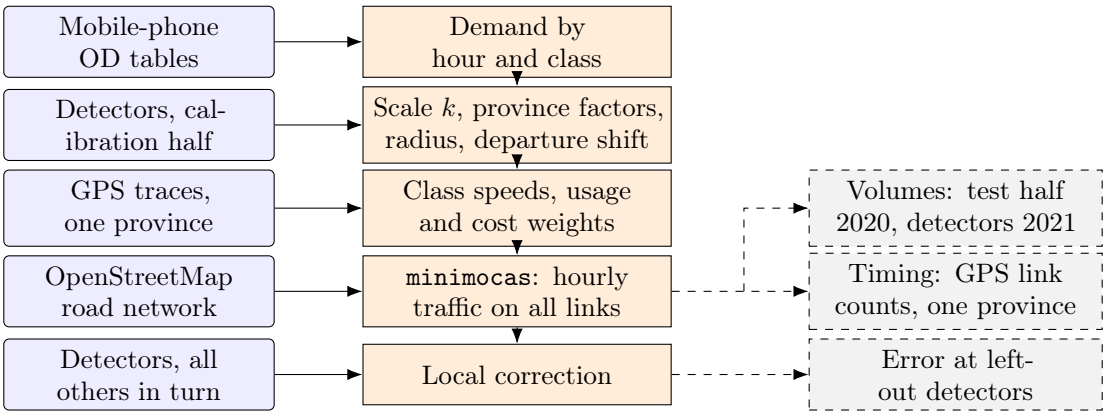


Figure 1: **Pipeline of the simulator.** Read from left to right: each input feeds one step of the model, and the steps together produce hourly traffic on all links, which is then compared with data on the right. Blue: input data. Orange: steps of the model. Grey, dashed: data used to assess the output, which do not set the values of the steps they assess.

2. Data

The road network comes from OpenStreetMap [52, 53], whose road coverage has been estimated to exceed 95% in most European countries [54]. It has 3,925 links and 2,548 nodes, with their length, direction and turn restrictions, and each link has an OpenStreetMap road class (Fig. 2a).

For the demand, the operator TIM provided hourly activity between municipalities for January–March 2020. We used 13 January–23 February, which excludes the holidays of early January and the restrictions for COVID-19 that began in the region on 23 February, and averaged it separately for each day of the week: each simulated day uses the mean of its own day of the week, so a Tuesday is simulated with the mean of the six Tuesdays of the period. The model keeps the relations among the 50 most active municipalities and between them and the provinces of neighbouring regions. When one end of a trip, or both, is outside this set, the model draws the missing end at random. We left out trips within each of the 50 municipalities, because adding them did not reduce the spread of the errors on the calibration detectors (Supplementary Section S2). Trips within the other municipalities stay in the demand, with both ends drawn at random.

So only part of the recorded activity enters the model as a relation. We give the figures for the mean Monday; those for the mean Saturday are similar (Supplementary Section S8). The Monday table holds 3.15 million trips per day, in three parts. 18% have the code 000000000 at one or both ends; this code identifies no place, and we remove these trips. 43% are within a municipality: those within the 50 municipalities (38% of the table) are left out, and those within the other municipalities (5%) enter with both ends drawn at random. The remaining 39% are between municipalities: 28% of them keep their relation, and 72% have a recorded relation that the model does not use and enter with their end in the selected set, if they have one, and the other end drawn at random. The result is 1.99 million agents on a Monday and 1.61 million on a Saturday, after scaling by k and the province factors. Of the Monday agents, 28% have a known origin and destination and 72% have at least one random end (Fig. 3; Supplementary Section S1). The configuration generator we inherited counted the trips with neighbouring provinces a second time among the agents with random ends (4.4% of the agents on a Monday before scaling), and we corrected this (Supplementary Section S2).

Since the demand depends only on the day of the week, the fifteen days of the two campaigns need one simulation for each day of the week, plus a second Monday that checks that a run is reproducible. The 2021 days therefore use demand measured in early 2020. The hour labels in the tables refer to local time at the start of the hour (Supplementary Section S7).

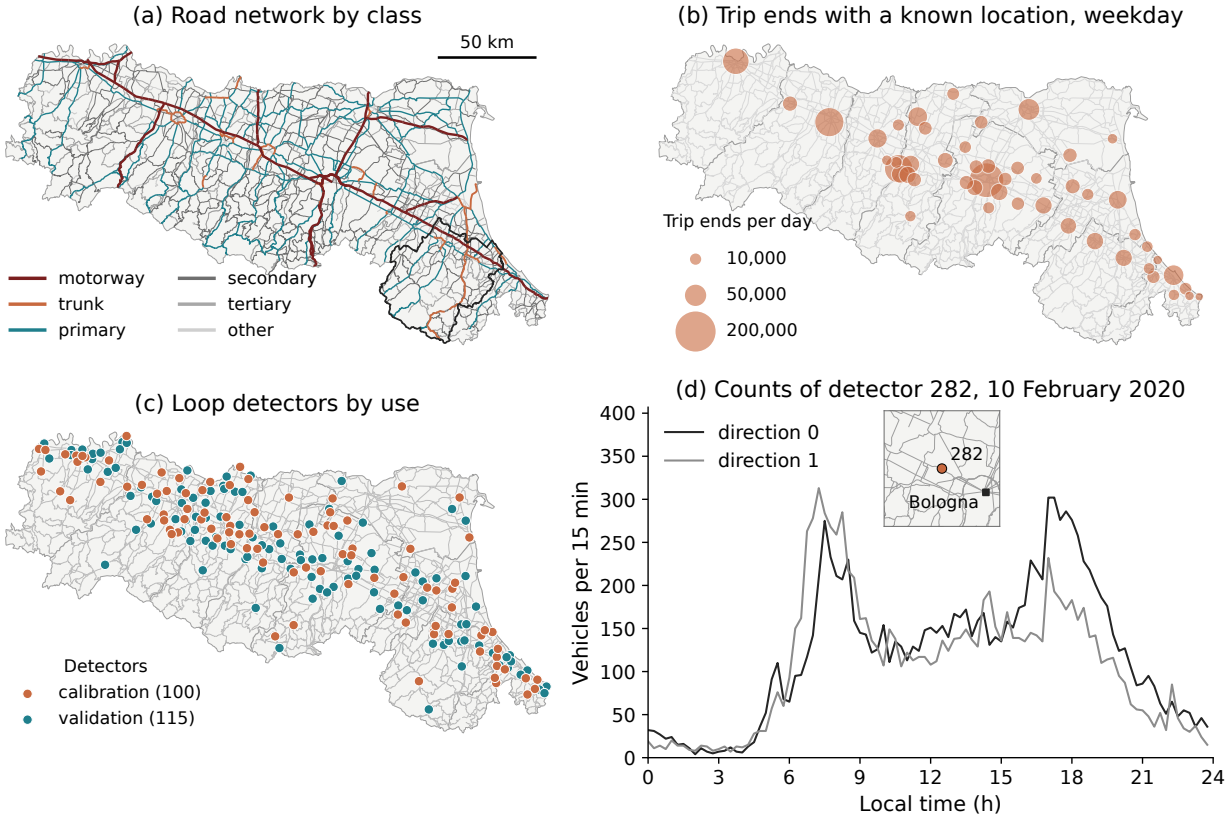


Figure 2: **The region and the data that enter the model. (a)** Road network coloured by OpenStreetMap road class; the province of Forlì–Cesena, where GPS traces inform route choice, is outlined. **(b)** Daily trip ends with a known location in the weekday demand, by municipality: a larger circle means more trips, and the demand concentrates along the central axis of the region and on the coast. **(c)** Loop detectors of the regional MTS system placed on the network, by their use for daily volumes: the calibration half of February 2020 (100; 60 of them also enter the 2021 tests) and validation only (115: the 97 of the test half and 18 used only in 2021). The other 77 detectors of the network have no complete counts in either campaign and are not shown. **(d)** Counts of detector 282 near Bologna on 10 February 2020, 15-min resolution, local time. Road network © OpenStreetMap contributors.

The regional MTS system counts vehicles every 15 min in both directions, and 292 of its detectors lie on links of the network (Fig. 2c,d). A detector is used on a given day only if both of its directions are complete. This filter favours busy roads, but relaxing it hardly changes the errors (Supplementary Section S7).

Viasat provided hourly probe counts per matched link for two campaigns, 10–17 February 2020 (six weekdays and a weekend) and 28 June–4 July 2021 (five weekdays and a weekend), and the raw traces of private vehicles in the province of Forlì–Cesena. We use the link counts

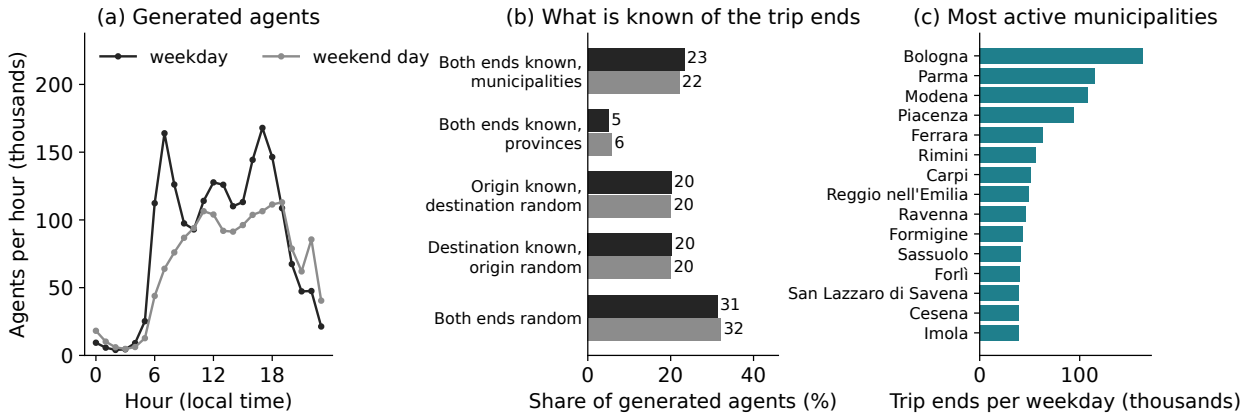


Figure 3: **Simulated demand. (a)** Agents generated per hour for a weekday and a weekend day, after scaling by k and the province factors: weekdays show morning and evening peaks, and weekends a single broad rise through the day. **(b)** Shares of agents by what is known of their trip ends; "provinces" are relations with provinces of neighbouring regions. More than a quarter of agents have both ends known, and the largest group has both ends drawn at random. **(c)** Municipalities with the most trip ends on a weekday.

of that province only (Supplementary Section S5). Probes were not used to generate or scale the demand, or in the detector correction.

Phone data were provided as anonymized counts aggregated in space and time, and we accessed no individual records. GPS data were provided as pseudonymized vehicle records and used only to compute link counts, speeds and road-class aggregates.

3. Methods

3.1. Simulation and route choice

The simulator moves agents in steps of 10 s, each at the speed of its link. This speed falls as the link becomes denser, down to a critical speed. Agents with a known origin and destination follow best paths computed with Dijkstra's algorithm [55], which respect turn restrictions. Agents with a random end instead choose the next link at each node, with a probability proportional to a transition weight, and travel for a time drawn between 900 and 5,600 s. Knowing their destination determines where the trip starts, but not the route they take. For each hour and direction the simulator records how many distinct agents used each link, while virtual counters at the detector locations record crossings. Each run starts empty at midnight, and a given random seed always gives the same run. Traffic is not assigned to an equilibrium [26], and the model does not reproduce the congestion phenomena of microscopic traffic-flow models [56, 57, 58].

To inform route choice, we matched GPS fixes of private vehicles in Forlì–Cesena (February

2020) to the road network. For each road class we took the median off-peak speed and the median daily number of probes per link (Supplementary Section S3), and applied both to all links of that class in the region. The speeds seem stable over time: on the same roads, those of June 2021 have a rank correlation of 0.98 with those of February 2020. The probe numbers set the transition weights of agents with a random end. They also enter the best-path cost of link i,

$$w_i = L_i \left[\alpha + \alpha_s \frac{v_0}{v_i} + (1 - \alpha - \alpha_s) \frac{1/c_i - 1/c_{\max}}{1/c_{\min} - 1/c_{\max}} \right], \tag{1}$$

where L_i is its length, v_i its speed, $v_0 = 20$ km/h a reference speed of the simulator that makes the speed term dimensionless (only the product $\alpha_s v_0$ enters, so v_0 only sets the scale of α_s), and c_i the usage weight of its class, so that busier road classes cost less. This cost describes an empirical preference by road class, and it is not a behavioural model. We chose the usage-weighted form among shortest-distance, travel-time and mixed alternatives because of its rank correlation between simulated daily volumes and GPS counts in Forlì–Cesena, without using detectors. The weights $\alpha = 0.5$ and $\alpha_s = 0.25$ come from a discrete grid compared on the calibration detectors; they are chosen values, not estimates (Supplementary Section S3).

3.2. Implementation, verification and computational cost

The simulator, `minimocas`, is written in C++14. Its inputs are the road graph with turn restrictions, the speed and usage weight of each road class, the trip sources with their creation rate in each 15-min interval, and the detector positions. Before the run it computes a best-path matrix for each class of agents with Dijkstra's algorithm, in parallel over nodes with OpenMP. Agents are then moved one after the other in each 10-s step, without parallelism, from 00:00 to 23:45 (8,550 steps). Outputs are the number of distinct agents per link, hour and direction, and the crossings of the virtual counters every 15 min.

We verified the implementation in three ways. A given random seed always gives the same run: the second simulated Monday of the 2020 campaign is identical to the first. After a defect in a diagnostic output was corrected, the engine reproduced the link flows of the final Monday byte for byte. The configuration generator, rewritten because the original was not available, writes the configurations only after it has reproduced the creation rates of the

inherited ones for the same period. Between four random seeds, the volume ratio and the Spearman correlation on held-out detectors change by at most 0.002 and the dispersion by at most 0.006 (Supplementary Section S8), well below the differences we interpret.

On a desktop computer (AMD Ryzen 3 3200G, four cores), one simulated Monday with 1.99 million agents took 7.1 min of wall-clock time, 6.5 min of which to move the agents, with a peak memory of 2.5 GB; it used about one core, with two OpenMP threads for the best paths. Its link flows and counter crossings were identical, byte for byte, to those of the final run used in this paper. Simulated days are independent, so the days of a campaign run in parallel as separate processes. We did not measure how the cost grows with the size of the network or the number of agents.

3.3. Demand scaling and local correction

Detectors enter the model in two ways. Half of them set the overall level of the demand, with one factor for each day type and one for each province. The counts of each day then correct the simulation locally, near each detector.

For the first use, we split the detectors with complete data in February 2020 into a calibration half (100 detectors) and a test half (97), with the same share of each province in both. Only the calibration half sets values; the test half and the 2021 campaign are kept for the tests.

The OD tables describe one operator's customers, not all traffic, so we multiply the number of agents by a factor k. This factor combines coverage, vehicle occupancy and the effect of random trip ends, and has no physical meaning of its own. In particular, it trades off against trip length: simulated agents travel about 40 km on a Monday, while the distances between municipalities in the tables average 12 km in a straight line (Supplementary Section S8). We compare a Monday simulation with the mean weekday counts and a Saturday simulation with the mean weekend counts, and for each we choose the k that minimizes the mean squared $\log_{10}$(simulated/observed) of daily volumes, on a grid from 0.9 to 1.8 (Supplementary Section S2).

The same loss chose the radius within which trip ends are drawn, among 15, 20, 25 and 30 km. Random ends lie within 25 km of the known end, and the ends of a known relation within 25 km of its two municipalities, weighted by population, so a relation fixes the area of

a trip and not its exact ends. Origins are also spread around their source point. Departures are 45 min earlier than the OD label: this reduced the hourly error on the Monday but not on the Saturday, and we apply it to both (Supplementary Section S2).

Even with the level set, the error still varied strongly from road to road. We measure this with the *dispersion*, the standard deviation of $\log_{10}$(simulated/observed) of daily volumes across detectors. On the calibration half, 38% of its variance was between provinces (Section 4.3), so we tested one more factor per province. Each trip source with a known location is multiplied by the geometric mean of the factors of its ends inside the region. Sources without a location and ends outside the region keep a factor of one, and k does not change. The factor of a province starts from one and is multiplied by 10^{-m}, where m is the median $\log_{10}$(simulated/observed) of its calibration detectors over both day types. We do this for two rounds. We fixed the rule for adopting the factors before the runs: the 95% paired bootstrap interval of the change in dispersion had to be below zero on the test half and, only if it was, on the detectors of 2021. The factors add nine parameters, and their values come from the calibration half only.

For the second use, where there is a detector, we can replace the simulated count with the observed one and spread the difference to nearby roads. Let $u_i(h)$ be the simulated count on the directed link of detector i in hour h and $o_i(h)$ the observed count in that direction; detectors on the same directed link are averaged. The increment at the detector link is

$$\delta_i(h) = \begin{cases} o_i(h) - u_i(h) & \text{if } u_i(h) \geq 1, \\ 0 & \text{otherwise,} \end{cases} \tag{2}$$

so the corrected count $\widetilde{u}_i(h) = u_i(h) + \delta_i(h)$ equals the observed one wherever the simulation has traffic at the detector. From there the increment spreads downstream and upstream. Each adjacent link receives the increment of the previous link, multiplied by its own simulated share of the flows leaving (downstream) or entering (upstream) the common node, $\delta_{(n_2,n)} = \delta_{(n_1,n_2)}\, u_{(n_2,n)} / \sum_m u_{(n_2,m)}$. The spreading does not enter other detector links. It stops when the increment falls below one vehicle, after three links along a path, at a forbidden turn or at a closed link. Any negative count is set to zero.

3.4. Evaluation metrics and tests

To assess timing, we compare on each GPS-matched link the hourly shares $p_i(h)$ and $q_i(h)$ of the reconstructed and GPS counts over the campaign days, both directions together,

$$\mathrm{MAE}_i = \frac{100}{24} \sum_{h=0}^{23} |p_i(h) - q_i(h)|, \tag{3}$$

in percentage points (pp), against three benchmarks: the OD profile of the demand, the mean profile of the detectors and a flat profile.

Daily volumes are compared at detectors with complete data, both directions, by the ratio of total simulated to total observed volume, the Spearman rank correlation, the median absolute $\log_{10}$ ratio and the share of detectors within a factor of two, with intervals from a paired bootstrap over detectors. The predictors without simulation take the level from the median volume of other detectors of the same road class and the profile from their mean profile. For the test half they are built on the calibration half, and for the leave-one-out test on the other detectors of the same day.

In the leave-one-out test, we leave out all detectors of a link in turn and compute the error $\sum |\widetilde{u}(h) - o(h)| / \sum o(h)$ of the corrected reconstruction on that link, over hours, directions and days. The depth of the correction, the number of links it spreads along, is the smallest whose error reduction in 5-fold cross-validation over 2020 is within one bootstrap standard error of the best (Supplementary Section S7).

The final tests use the raw simulation on the test half of 2020 and on all detectors of 2021, and each supports only the property it tests. For the level, the volume ratio is equivalent to one within 0.80–1.25 if its 90% bootstrap intervals over detectors and over provinces both lie within that margin, as in two one-sided tests; since we chose the margin, we also report the smallest one that passes. For the ranking of roads, we compare the Spearman correlation with permutations of the simulated volumes, among all detectors and only within road class; the second test looks only at what road class alone does not explain. For timing, the hourly-share error on the links of Forlì–Cesena should be below that of the flat profile (one-sided Wilcoxon signed-rank test). The p values of the ranking and timing tests are Holm-adjusted over the six tests, three in each campaign.

These data are held out but not blind. They decided whether to adopt the province factors. The 2021 set includes 60 calibration detectors, so we also report it without them. The removal of the code 000000000 and the travel-time term in the cost were decided after earlier versions had been compared on them. Numerical values were always set on the calibration half (Table 1), but the results are held out rather than pre-registered.

The experiments on the OD relations followed protocols written before their runs, are exploratory and do not change the model; they are described with their results in Supplementary Section S9.

4. Results

4.1. Level and timing of traffic

We first ask whether the simulator puts the right amount of traffic on the network. Scaling the demand by day type, with $k = 1.5$ on weekdays and $k = 1.3$ on weekends, brings simulated and observed volume on the calibration detectors within 7% and 10%, and within 1% and 4% with the province factors. A single factor of 1.4, the best common value, would leave weekdays 1% too low and weekends 18% too high (Supplementary Fig. S1). The selected values remain the minima of the grid with province factors, and subsets of 50 calibration detectors choose the same value or a neighbour in nearly all cases (Supplementary Section S8). Regional differences remain after this step, and the province factors correct them. Figure 4a shows that the correction is far from uniform: demand is raised in the north-west, in Modena and in Ferrara and lowered along the east and in Reggio Emilia. We treat these factors as empirical, not as measured differences in phone coverage or car use.

Figures 4b and 4c show the outcome at detectors that did not set these values, or set them only in part in 2021. In both campaigns the cloud of points follows the diagonal: the overall level is right and busy roads tend to be simulated as busy, while the width of the cloud shows the error at single roads. Table 2 shows what the province factors contribute. Their gain is clearest at the typical detector, where the median error falls by about a fifth, and the rule for adopting them was met in both data sets. Factors estimated on 25 or 50 calibration detectors mostly give a dispersion close to that of the 100 used here, although the value for a single province varies strongly between subsets, so the factors should be read as effective corrections

and not as precise estimates (Supplementary Section S8). All later results use the model with the factors.

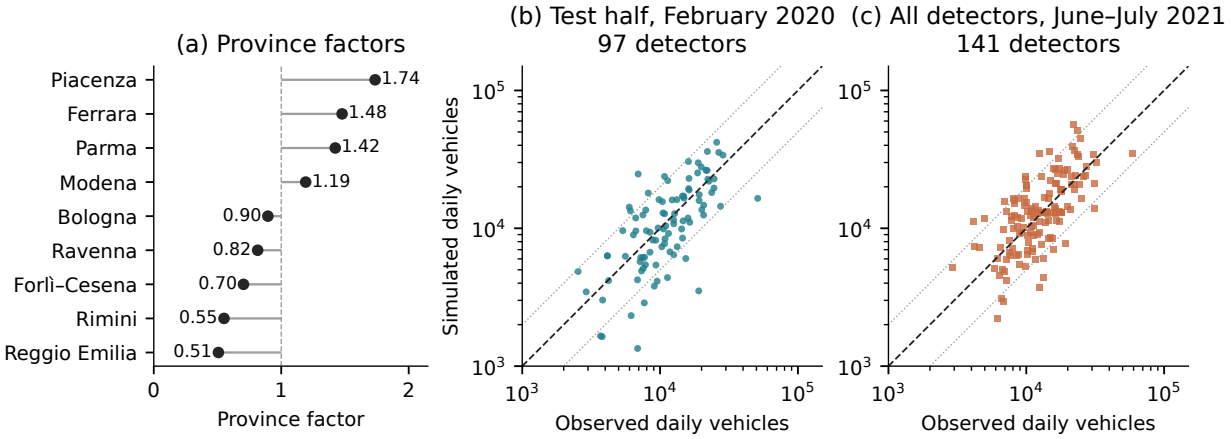


Figure 4: **The simulator reaches the overall level of traffic. (a)** Factor applied to the demand of each province; values above one (dashed line) raise the demand. The factors were set on the calibration half of the detectors. **(b,c)** Daily volume at each detector, simulated against observed, raw simulation before any local correction, on logarithmic axes. A point on the dashed line would be a perfect estimate, and points between the dotted lines are within a factor of two. A cloud centred on the dashed line means that the level is right overall, and its width shows the error at single roads. Panel (b) uses detectors kept out of all calibration; panel (c) uses a later season and includes 60 calibration detectors. One detector in each panel has no simulated traffic and is not drawn.

Table 2: Effect of the province factors, without and with them. Lower dispersion and median error, higher Spearman correlation and a volume ratio closer to one are better; an interval that excludes zero marks a change the data can establish. Dispersion: standard deviation of $\log_{10}$(simulated/observed) of daily volumes across detectors; the test-half values include detector 153, which has no simulated traffic. Median error: median absolute $\log_{10}$ ratio per detector and day type. Intervals: 95% paired bootstrap of the change.

	Calibration half		Test half, 2020		2021	
Measure	without	with	without	with	without	with
Dispersion	0.222	0.185	0.500	0.489	0.415	0.399
change, interval	$[-0.052, -0.022]$		$[-0.034, -0.003]$		$[-0.037, -0.008]$	
Spearman correlation	0.66	0.74	0.63	0.66	0.63	0.66
change, interval	$[0.02, 0.13]$		$[-0.01, 0.08]$		$[-0.01, 0.07]$	
Median error	0.151	0.101	0.180	0.148	0.177	0.142
Volume ratio	1.08	1.02	1.06	1.01	1.19	1.10

With the level set, we turn to when traffic occurs, using GPS as a view of the daily rhythm on the matched roads of Forlì–Cesena. Figure 5a,b compares the simulation with two simple alternatives: the hourly profile of the demand itself and the average profile of the detectors. All three are far better than a flat profile, and the curves are close: most of the daily rhythm is already in the demand. The simulation improves on the demand profile, and after correction it stays slightly behind the detector average in both campaigns (Table 3). Weekend timing is harder than weekday timing for both the simulation and the detector average (Supplementary

Fig. S4). GPS traces of this province also set the route weights, so this comparison is not fully independent of route choice, although route choice hardly changes the profiles (Supplementary Section S3).

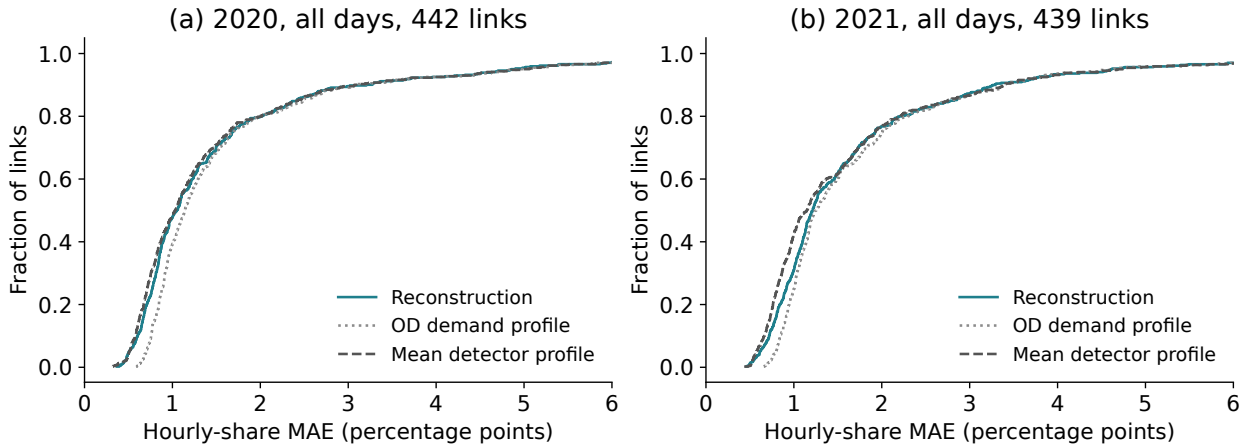


Figure 5: **The timing of traffic follows GPS on nearly all links.** For each GPS-matched link we compute the error of its hourly shares, the mean absolute difference between the share of daily traffic in each hour in the estimate and in GPS. Each curve shows the fraction of links whose error is below the value on the horizontal axis, so a curve further up and to the left is better. Reconstruction: simulation after local correction. **(a,b)** All days, against the OD demand profile and the mean detector profile, in 2020 and 2021. The curves are close, and a minority of links has large errors for all three. Curves continue beyond the 6 pp limit of the axis.

Table 3: How well each estimate reproduces the daily rhythm: median hourly-share error (pp) on the GPS links of Forlì–Cesena, all days of each campaign; lower is better. Raw: simulation. Corrected: simulation after the local correction. Mean detector, OD and flat: profiles without simulation, as benchmarks.

Campaign	Links	Raw	Corrected	Mean detector	OD	Flat
2020	442	1.08	1.05	1.03	1.14	2.72
2021	439	1.21	1.21	1.15	1.25	2.60

Detectors count all vehicles, so they let us look at the actual flow through the day. Figure 6 shows four test detectors for each day type, one from the middle of each quartile of the 15-min error, and where they lie in the region. On weekdays the simulation follows both peaks and the midday plateau at all four detectors, also where the level is wrong (profile correlations 0.93–0.97), so the error comes mostly from the level. On weekends it follows the general shape less closely, starting too low at night and running too high late in the evening. Table 4 gives the picture over all test detectors. The hourly error is about 40%, and it would fall by about two thirds on weekdays if each detector were given its observed daily level, so at a single road the difference is mostly how much traffic there is, not when. All detectors share a similar daily pattern, and the plain average of the calibration detectors has a higher profile correlation with each test detector than the simulation has. The simulation, however, ranks the roads

clearly better than a predictor based on road class.

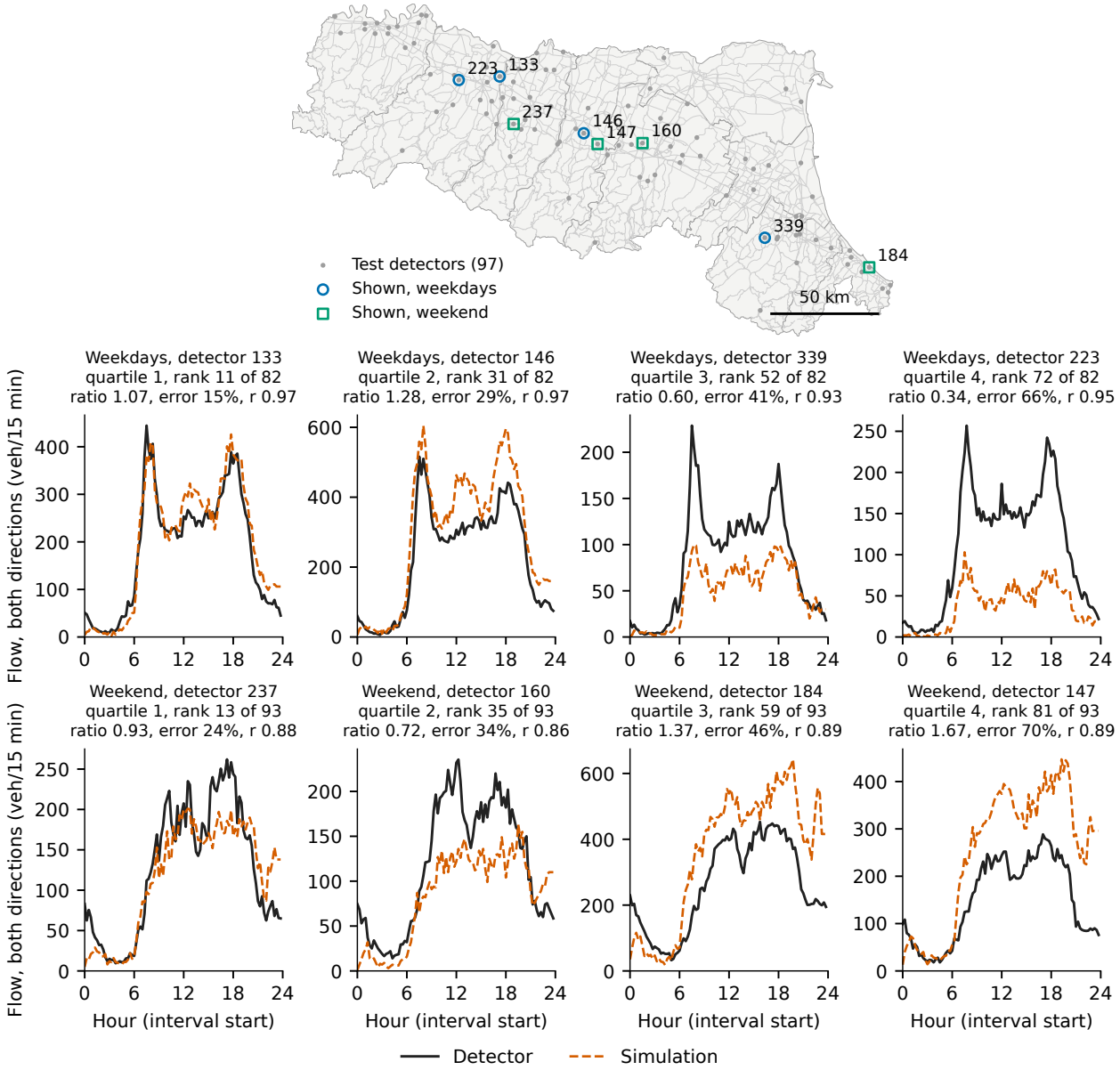


Figure 6: **Daily profiles at test detectors across the range of the error.** Top: road graph, test detectors and the detectors shown below, by day type. Below: one test detector from the middle of each quartile of the 15-min error, from the lowest (left) to the highest (right). Mean observed flow (solid) and raw simulation before any correction (dashed), both directions, February 2020, 15-min resolution, for weekdays (top) and weekends (bottom). The title of each panel gives the quartile and rank among test detectors, the ratio of simulated to observed volume, the 15-min error and the correlation of the two curves. The median detector has a 15-min error of 33% on weekdays and 38% on weekends.

4.2. Daily volumes and local correction

Since most of the difference at detectors is about level, we now look at daily volumes (Table 5, top). The results of the two campaigns are similar: the total volume is close to the observed one, the ranking of roads is good and more than four detectors in five are within a factor of two. The level in 2021 is reached with demand measured in early 2020 and a scale

Table 4: Raw simulation against a predictor without simulation, test detectors of February 2020. A volume ratio close to one, lower errors and higher correlations are better. The predictor, built on the calibration detectors, is the median volume of the same road class times the mean profile. The province factors were also set on those detectors. Volumes and errors come from the virtual counters, both directions; these counters are about 8% below the distinct-agent counts on which k is set. Intervals: 95% bootstrap of the difference, simulation minus predictor. Peaks, simulated against observed: weekdays 8:00 and 18:00 against 7:30 and 17:30; weekend evening 19:15 against 17:30.

	Weekdays (82 detectors)			Weekends (93 detectors)		
Measure	Sim.	Pred.	Interval	Sim.	Pred.	Interval
Volume ratio	0.93	0.92		0.93	0.88	
Median $\lvert\log_{10}\rvert$ ratio, daily volume	0.151	0.137	$[-0.05, +0.05]$	0.145	0.157	$[-0.06, +0.03]$
Within a factor of two (%)	83	89		81	88	
Spearman, daily volume	0.67	0.41	$[0.06, 0.46]$	0.68	0.48	$[0.02, 0.36]$
Hourly error (%)	37.9	40.1	$[-11.1, +6.6]$	43.0	41.1	$[-6.0, +9.5]$
Hourly error with known level (%)	12.9	7.2		22.4	8.2	
Median profile correlation	0.96	0.99	$[-0.04, -0.03]$	0.86	0.98	$[-0.12, -0.10]$

fixed on winter counts, so it should not be read as a test of demand in that season. Without the 60 calibration detectors, the 81 remaining detectors of 2021 give a volume ratio of 1.11 [0.99, 1.24], a Spearman correlation of 0.65 [0.50, 0.76] and 81% [73, 90] within a factor of two. Road class adds a further pattern (Fig. 7): the simulation underestimates secondary roads (median ratio 0.78 in both campaigns) and tends to overestimate primary roads slightly (1.05 and 1.12), while trunk and tertiary roads have too few detectors for a clear value. No detector lies on a motorway, so the level on motorways, 709 km of the network, is not tested.

The contribution of GPS appears when route choice is changed under the same demand (Table 5, bottom, and Fig. 9a). At detectors outside the province where route choice was set, GPS-informed routing ranks roads better than shortest-distance routing in both campaigns, while the typical error does not change measurably. This gain comes mainly from the use of each road class, and GPS speeds alone add little (Supplementary Section S8). Repeating the selection on the final model confirms the chosen form. A simple predictor gives a useful comparison: road class and the mean profile, with no simulation, give a similar error on daily volumes, and the simulation does better in ranking the detectors (Table 4).

All results so far describe the raw simulation. We now add the local correction. To test it fairly, we leave out each detector in turn and ask the correction, which uses the other detectors, to predict its count. The correction removes 2.4 percentage points of the error in 2020 and 6.0 in 2021 at detectors it has not seen (Table 6), and the spread between detectors remains larger than the gain. About three detectors in five improve and about one in four

Table 5: Daily volumes at held-out detectors, raw simulation, with 95% bootstrap intervals. A volume ratio of one means that the total is exact, a Spearman correlation of one that roads are ranked exactly, and a median $|\log_{10}|$ ratio of 0.15 that the typical road is within about 40% of its count. Top: all detectors of each set. Bottom: GPS-informed against shortest-distance routing under the same demand, at test detectors outside Forlì–Cesena; intervals are for the paired difference.

Measure	Test half, 2020 (97 detectors)	2021 (141 detectors)
Volume ratio	1.00 [0.89, 1.11]	1.10 [1.01, 1.20]
Spearman correlation	0.69 [0.56, 0.79]	0.66 [0.54, 0.74]
Median $\lvert\log_{10}\rvert$ ratio	0.148 [0.113, 0.176]	0.142 [0.118, 0.174]
Within a factor of two (%)	81 [73, 90]	84 [78, 90]
	Test half, outside FC (87)	2021, outside FC (1
Spearman, shortest distance to GPS	0.63 to 0.71, +0.08 [0.01, 0.17]	0.54 to 0.64, +0.10
Median $\lvert\log_{10}\rvert$ ratio, shortest distance to GPS	0.151 to 0.147, 0.00 [−0.04, 0.03]	0.152 to 0.141, −0.0

gets worse. The 2020 set includes the calibration detectors and the depth of the correction was selected on 2020, so 2021 is the estimate free of that selection (Supplementary Section S4). With a fixed 20% held out on single days, the gain grows with the number of detectors, from 1.4 to 3.1 pp on the winter day and from −0.1 to 1.4 pp on the summer day (Supplementary Section S8).

Again we compare with the predictor built on the other detectors of the same day (Table 6 and Fig. 9c). The corrected reconstruction cannot be told apart from it, and each is better at about half of the detectors. The simulation, in addition, gives a complete and consistent field over all links of the network.

Table 6: Local correction at left-out detectors, depth 3: each detector is predicted by a correction that does not use it, so a lower corrected error is a real gain at a road the correction has not seen. Hourly error pooled over hours, directions and days. Predictor: road class and mean profile from the other detectors of the same day. Intervals: 95% bootstrap.

Measure	2020 (152 detectors)	2021 (141 detectors)
Hourly error, raw (%)	42.2	47.4
Hourly error, corrected (%)	39.8	41.4
Hourly error, predictor (%)	36.9	37.6
Reduction by the correction (pp)	2.4 [0.2, 4.8]	6.0 [2.5, 10.3]
Raw minus predictor (pp)	[−1.3, +11.6]	[+1.7, +18.9]
Corrected minus predictor (pp)	[−3.0, +8.9]	[−3.1, +10.8]
Median detector, raw to corrected (%)	35.9 to 33.3	39.4 to 34.8
Detectors improved / worsened (%)	60 / 26	62 / 23
Detectors better than the predictor (%)	47	50
Spearman of volumes, raw to corrected	0.72 to 0.72	0.66 to 0.69

With the correction we can draw the full traffic field (Fig. 8). The motorway axis and the roads around the main cities carry most traffic, and the Apennine roads in the south carry

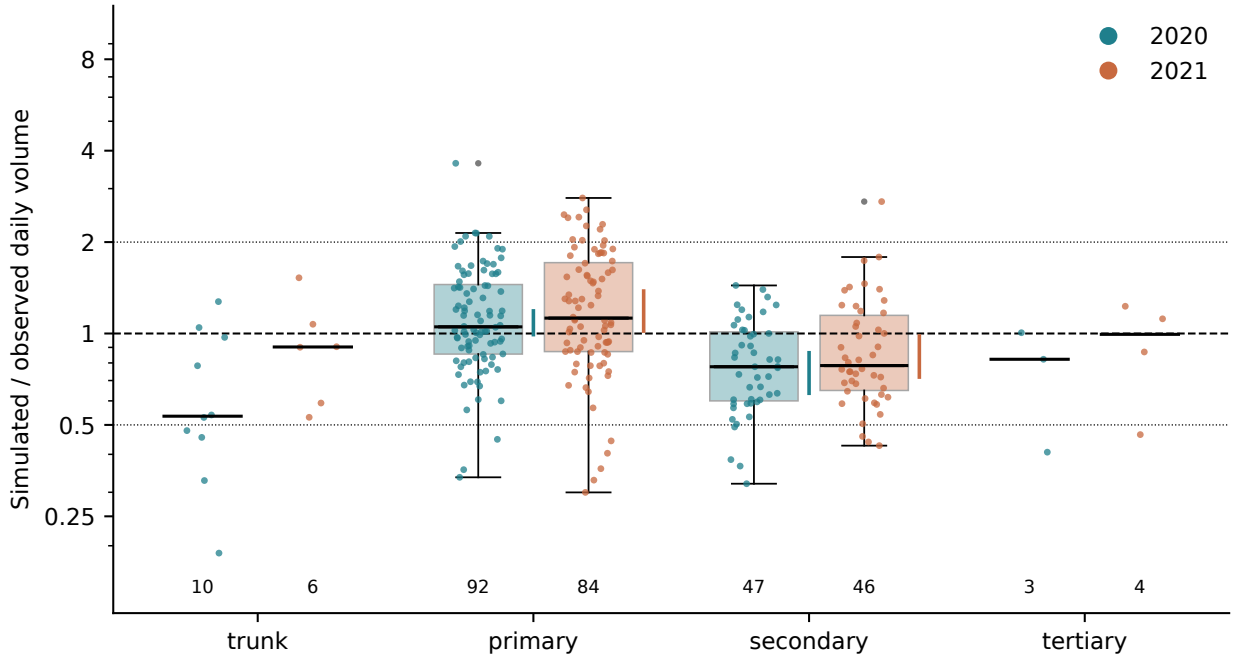


Figure 7: **Simulated volume by road class.** Ratio of simulated to observed daily volume, raw simulation, on the detectors with complete days of the leave-one-out test. Each point is a detector, and the number of detectors is below each group; 106 detectors appear in both campaigns. Black bars: medians. Primary and secondary roads also show quartiles and whiskers at 1.5 interquartile ranges, and the coloured bars beside them are 95% bootstrap intervals of the median. The dashed line marks perfect agreement and the dotted lines a factor of two. Trunk and tertiary roads have few detectors, and motorways have none. One detector with zero simulated traffic in 2021 is not shown.

little. The inset shows the best case of the test: with detector 349 left out, the reconstruction on its road follows the daily curve and misses the daily total by about 3%. This is what the simulator can achieve where it works best (Table 6 gives all detectors). Figure 9 collects the gains and differences discussed so far, together with the OD relations.

We tested the OD relations in detail in Supplementary Section S9. Removing them lowers the ranking of roads on the test half, but this also removes the regional pattern of the province factors. With that pattern kept, no gain in ranking can be established, and the spread of errors is larger with the relations. Destinations rearranged at random make trips far too long and tend to raise the spread, while destinations rearranged at the same distances cannot be told apart from the real ones. The value of specific origin–destination pairings could not be determined from link-count observations with the statistical power available in this study: in the registered test the power was at most 0.11, far below the 80% it required.

4.3. Structure of the remaining error

With the regional level, the ranking and the timing in place, we can ask what kind of error remains at each road. Figure 10a,b shows that it is not noise but a stable property of the road.

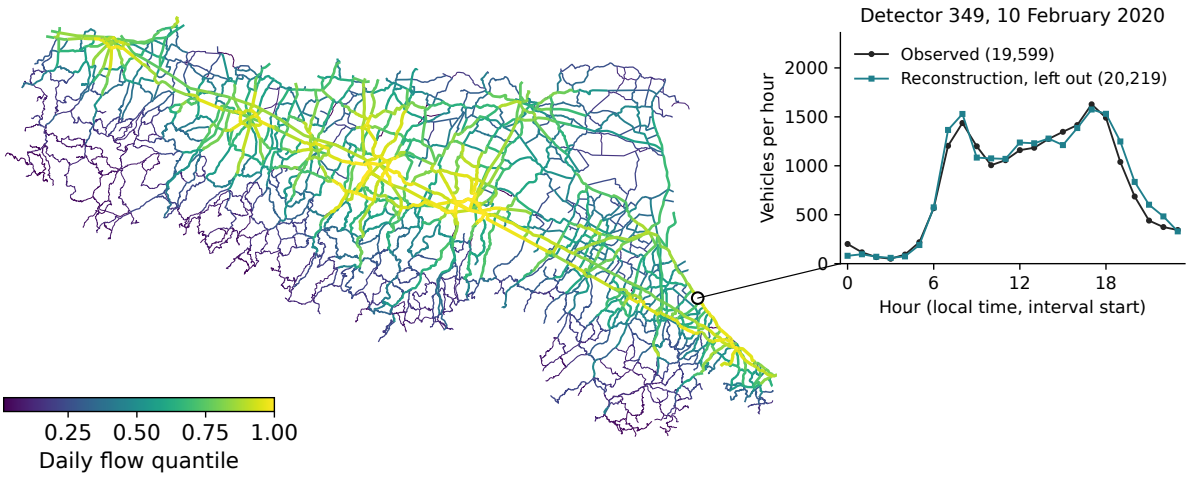


Figure 8: **Reconstructed traffic field.** The simulation and the local correction give traffic on every link, including roads without detectors. Map: reconstructed daily flow on each link, 10 February 2020, with all detectors used. Colour and line width show the quantile of the flow, so yellow marks the busiest roads of the region. Inset: the best case among the 61 detectors complete on that day, detector 349 (circle on the map): hourly counts, both directions, and the reconstruction on its link when that detector is left out of the correction and all others are used; daily totals in parentheses. The error is 9%, against a median of 35% over the 61 detectors.

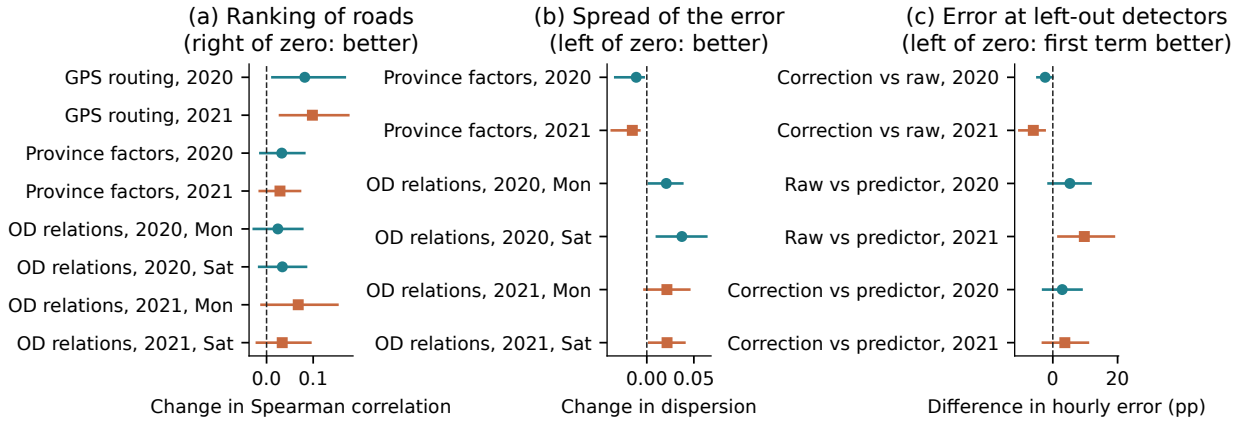


Figure 9: **The marginal information of each source.** Each row is one comparison on data not used to set the values: the marker is the estimated change and the bar its 95% bootstrap interval. A bar that crosses the dashed zero line means that the data cannot establish a change. Blue circles: 2020; orange squares: 2021. For the OD relations, the regional pattern of the factors is kept in the simulation without relations (Supplementary Section S9). **(a)** Change in the rank correlation between simulated and observed volumes when the source is added. GPS routing helps in both campaigns; no interval of the OD relations clears zero. **(b)** Change in dispersion, the spread of the error between roads. The province factors reduce it, and the OD relations do not. **(c)** Differences in hourly error at left-out detectors. The correction beats the raw simulation, but not the predictor without simulation.

A road that the simulator overestimates on weekdays is overestimated by about the same amount on weekends ($r = 0.95$), and the same holds between winter 2020 and summer 2021 ($r = 0.98$). The day-to-day variation of the counts is about ten times smaller than this error. Part of the stability between seasons is expected, since both campaigns use the same demand; it also means that the counts on each road changed little relative to the simulated ones.

Figures 10c and 10d show what the province factors did to this error. Without them, province was its largest known component and nearby detectors had similar errors. With them, that regional component has mostly gone and nearby errors are no more alike than chance would give. Together, province, road class and volume now explain about a quarter of the variance. Misplaced detectors do not explain the rest: comparing their coordinates with the road geometry, we flagged 5 of 292 as doubtful, and dispersion remains large without them (Supplementary Section S5). A further analysis, out of sample, shows that road class is the only group of road properties with a repeatable signal, whereas nearby population and activities, centrality, parallel roads and the density of OpenStreetMap roads not in the network do not explain the error in a way that carries over between provinces (Supplementary Section S8). Route weights from the GPS count of each link, instead of its class, did not give a clear gain at the detectors of Forlì–Cesena, on few detectors (Supplementary Section S8). What remains is a stable property of each road, which may reflect how demand reaches the road locally, and which these data cannot separate further. Its stability suggests that local information could address it, as the correction already does where detectors exist.

4.4. Formal tests and contribution of each source

Finally, we test each property on held-out data (Table 7). The overall level is equivalent to the observed one within the chosen margin, the ranking of roads exceeds chance even when road class is held fixed, and the timing beats a flat profile on nearly all links of Forlì–Cesena. A regression across detectors gives a slope close to one, with much of the variance left to the single-road error described above. The province interval of the level test is based on few provinces, each of which received its own factor, so it is only partly held out.

Table 8 collects the effect of each source on data not used to set it. Phone data give the timing and, with detector scaling, the level; GPS improves the ranking of roads; detectors also bring a local gain through the correction. The OD tables add the short-range structure

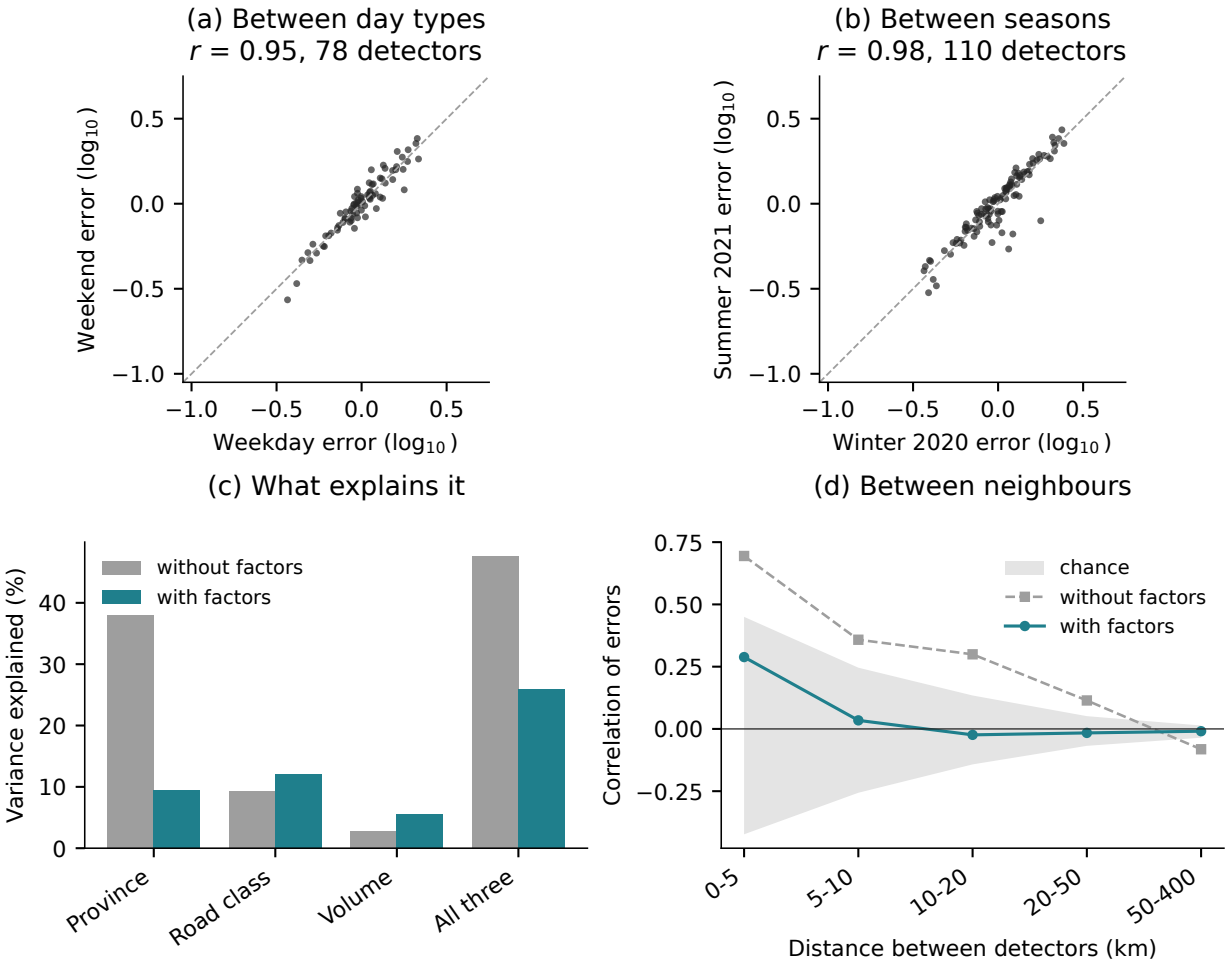


Figure 10: **The remaining error is stable and specific to each road.** The error is $\log_{10}$(simulated/observed) of daily volume, so zero is perfect and 0.3 is about a factor of two. **(a)** Error of each calibration detector on weekdays against weekends. **(b)** Error of each detector in winter 2020 against summer 2021. Points close to the dashed diagonal have the same error in both conditions. **(c)** Share of the variance of the error across calibration detectors explained by province, road class, observed volume and all three together, on the Monday, without and with province factors. **(d)** Correlation between the errors of pairs of detectors, by distance. The shaded band is the 95% range expected by chance, from permutations with the factors; a point inside it shows no detectable similarity between neighbours.

Table 7: Final tests on data not used to set the model values, raw simulation. The volume ratio passes if its interval lies within 0.80–1.25; the two Spearman tests ask whether the ranking of roads is better than chance, overall and within road class; the timing test whether the daily timing is better than a flat profile. Intervals for the volume ratio are 90%; for the slope, 95%. FC: Forlì–Cesena. The smallest margin of the volume ratio that passes in both campaigns is about 1.20. Holm-adjusted $p < 0.001$ for the two Spearman tests. In the timing test the simulation beats the flat profile on more than 99% of links. The 2021 set includes 60 calibration detectors.

Test	Test half, 2020 (97 detectors)	2021 (141 detectors)
Volume ratio, resampling detectors	1.00 [0.91, 1.09]: pass	1.10 [1.02, 1.18]: pass
Volume ratio, resampling provinces	[0.86, 1.13]: pass	[0.97, 1.20]: pass
Spearman above chance	0.69, $p < 0.001$	0.66, $p < 0.001$
Spearman within road class	$p < 0.001$	$p < 0.001$
Timing against flat, FC links (error, pp)	1.08 against 2.72, $p < 10^{-70}$	1.21 against 2.60, $p < 10^{-70}$
Slope, log–log volumes (R^2)	0.94 [0.72, 1.18] (0.19)	0.86 [0.70, 1.05] (0.17)

of trips; the value of specific pairings beyond it remains undetermined.

5. Discussion

We built a simulator of ordinary regional traffic from three data sources. With eleven demand-level parameters (two by day type, nine by province), it reaches the observed overall level and timing, it ranks roads well above chance, and each source has a measurable role. Detectors set the level and reduce the regional bias. GPS use by road class improves the ranking of roads. Phone tables give the daily timing and the short-range structure of trips. The correction brings a further local gain. The province factors are empirical: a factor of 1.74 or 0.51 cannot be read as a difference in phone coverage or car use without further evidence, and the factors would have to be estimated again for another period or region.

The OD tables need careful reading, because our tests answer narrower questions than "do OD data have value?". Destinations rearranged at random make trips far too long and tend to worsen the model, so the tables carry useful information on trip length. Random ends drawn near the known end supply much of the same information, which may be why removing the relations does not raise dispersion; this explanation is a hypothesis. The value of specific origin–destination pairings could not be determined from link-count observations with the statistical power available in this study. Just over a quarter of the trips between municipalities with known ends keep their relation, and link counts respond little to pairings:

Table 8: What each source adds, on data not used to set it. Each row gives a source, its job in the model and the change measured when it is added. Differences are in the sense of adding the source; negative errors are improvements. Intervals are 95% bootstrap.

Source	Role	Effect
Mobile-phone OD, total and timing	agents per interval	hourly-share error on FC links 1.08 and 1.21 pp against 2.72 and 2.60 flat; 0.06 and 0.06 pp lower than the OD profile (median paired difference); mean detector profile 1.03 and 1.15 pp
Mobile-phone OD, relations as used (28% of the trips between municipalities with known ends)	known trip ends	with the factors kept on both sides: Spearman +0.02 to +0.07, intervals include zero; dispersion +0.021 to +0.037 (worse), three of four intervals exclude zero. Against random pairs: dispersion −0.017 to −0.039, all four intervals exclude zero, two of them narrowly. Against pairs with the same distances: no interval excludes zero at 50 municipalities, one of four at 347; value of the pairings not determined (power at most 0.11)
Detectors, half	scale k by day type	test ratio 1.00; 1.10 in 2021
Detectors, half (nine values)	province factors	median $\lvert \log_{10} \rvert$ ratio per detector and day type 0.180 to 0.148 on the test half, 0.177 to 0.142 in 2021; dispersion −0.011 and −0.015 (intervals exclude zero)
GPS traces	route weights	Spearman +0.08 [0.01, 0.17] and +0.10 [0.03, 0.18] outside the province; median error not measurably different; profiles unchanged
Detectors, all	local correction	hourly error −2.4 pp [−4.8, −0.2] and −6.0 pp [−10.3, −2.5] at left-out detectors; profile error −0.01 and 0.00 pp on GPS links

at the 25-km radius of the model, the change at detectors is below the variation between seeds. This is not evidence that pairings have no value, and they may be useful in applications that need trips rather than link volumes. In the same way, a value of $k = 1.5$ cannot be read as the inverse of the operator's coverage.

The simulation ranks the test half better than a predictor without simulation, with a similar error at the detectors. Unlike that predictor, it also provides routes and traffic on roads without detectors. Detectors are not a random sample of roads.

Three findings may apply beyond this region, as hypotheses for other settings. Timing needs a benchmark, since a common daily profile already explains most of the hourly pattern. A few scale parameters can reach the overall level, so the error at single roads should be reported along with it. Removing a source can mislead when other parameters act through it, so its effect should be tested with those parameters kept, and with the marginals preserved. The values we report are marginal to this model and to the order in which we built it, and they are not an exchangeable decomposition. They also concern ordinary traffic.

None of the fitted values should be carried to another region or period: the demand scale, the province factors, the class speeds and usage, the radius and the departure shift would all have to be estimated again. What can be carried over is the procedure: one role per source, held-out data for each role, predictors without simulation as benchmarks, and the effect of each source measured with the parameters that act through it held fixed. Whether other regions show the same pattern of contributions is an open question.

Some boundaries of these results should be kept in mind. GPS comparison covers one province, which also informs route choice. Several routes per relation, shorter walks and a smaller radius improved the ranking of roads in exploratory tests but did not reduce dispersion (Supplementary Section S8), so the reach of the relations remains limited by the structure of the model. The experiments on the OD relations are exploratory and rest on one simulated Monday and Saturday per campaign. The test half and 2021 took part in the decision on the province factors and had been seen when the final configuration was fixed. Several settings (trip radius, cost weights, origin spreading) were chosen on the calibration half without a demonstrated gain on held-out detectors. The largest remaining error is the level at individual roads, so aggregate agreement should not be read as accuracy at a given road.

This paper establishes what the simulator does for ordinary traffic: regional level, ranking of roads and timing, and hourly volumes on roads without detectors that are consistent with the network. It can therefore serve as a computational basis for later work on road closures, demand perturbations and network resilience. We make no claim here about those applications: skill on ordinary traffic does not imply skill under a change, and they need their own assessment on observed changes.

6. Conclusions

We have built a regional traffic simulator from mobile-phone demand, detector counts and GPS, and measured what each source adds on data not used to set each step. The study shows that such a simulator can be validated on independent observations, and that the contribution of each input can be measured, when each source has one role and is tested on data it did not set. Scaling by day type and province gives the observed overall volume on held-out detectors and reduces the regional bias. Route weights based on GPS use improve the ranking of roads, and the local correction reduces the error at left-out detectors. The overall timing comes from the daily profile of the demand. The OD tables contribute the level, the timing and the short-range structure of trips; the value of specific pairings beyond that structure could not be determined from link counts with the power available. Tests on held-out data support the overall level, the ranking of roads and the timing. At single roads the error is stable and specific to each road, and the simulation is comparable to predictors based on road class and other detectors. A simulator of this type gives a consistent picture of ordinary traffic at the scale of the region, over all links of the network, and its use for scenarios remains future work that needs its own assessment on observed changes.

Supplementary material

The Supplementary Material describes the data layers and the demand classes (S1), the demand scale and the changes tested on the calibration detectors (S2), the simulation model and the route-choice variants (S3), the detector correction and its sensitivity to depth (S4), additional results (S5), the final configuration and the reproduction of all results (S6), further

details of data and methods (S7), the experiments on the final model (S8) and the value of the OD relations (S9).

Abbreviations

GPS, Global Positioning System; MAE, mean absolute error; MTS, regional traffic monitoring system of Emilia-Romagna; OD, origin–destination; pp, percentage points.

CRediT authorship contribution statement

Federico Bellisardi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Chiara Mizzi:** Software, Data curation. **Alessandro Fabbri:** Software, Data curation. **Armando Bazzani:** Conceptualization, Methodology, Supervision.

Declaration of competing interest

C.M. and A.F. are employed by Mindicity, which contributed to the development of the modelling framework. The other authors declare no competing interests.

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Data availability

The mobile-phone OD tables (TIM) and the Viasat GPS data are proprietary and were used under agreements that do not allow redistribution; requests for access should be addressed to the data providers. MTS loop-detector counts are published by Regione Emilia-Romagna under a CC BY licence [59]. Road classes are derived from OpenStreetMap (© OpenStreetMap contributors, ODbL). The road graph and the aggregate results behind the figures and tables are included in the code repository; link-level metrics derived from the GPS data are not distributed. The simulation configurations derive from the mobile-phone OD tables and are not distributed. The simulator `minimocas`, the detector correction and all analysis scripts are

available in the code repository [60]. Supplementary Section S6 lists the final configuration and the steps that reproduce the results.

Declaration of generative AI and AI-assisted technologies in the writing process

[To be completed by the authors.] During the preparation of this work the authors used [name of tool/service] in order to [reason]. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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References

[1] J. L. Toole, S. Colak, B. Sturt, L. P. Alexander, A. Evsukoff, M. C. González, The path most traveled: Travel demand estimation using big data resources, Transportation Research Part C: Emerging Technologies 58 (2015) 162–177. doi:10.1016/j.trc.2015.04.022.

[2] S. Çolak, A. Lima, M. C. González, Understanding congested travel in urban areas, Nature Communications 7 (1) (2016) 10793. doi:10.1038/ncomms10793.

[3] N. Grujić, S. Brdar, S. Osinga, G. J. Hofstede, I. N. Athanasiadis, M. Pljakić, N. Obrenović, M. Govedarica, V. Crnojević, Combining Telecom Data with Heterogeneous Data Sources for Traffic and Emission Assessments—An Agent-Based Approach, ISPRS International Journal of Geo-Information 11 (7) (2022) 366. doi:10.3390/ijgi11070366.

[4] M. C. González, C. A. Hidalgo, A.-L. Barabási, Understanding individual human mobility patterns, Nature 453 (7196) (2008) 779–782. doi:10.1038/nature06958.

[5] F. Calabrese, G. Di Lorenzo, L. Liu, C. Ratti, Estimating Origin-Destination Flows Using Mobile Phone Location Data, IEEE Pervasive Computing 10 (4) (2011) 36–44. doi:10.1109/mprv.2011.41.

[6] A. Wesolowski, N. Eagle, A. M. Noor, R. W. Snow, C. O. Buckee, The impact of biases in mobile phone ownership on estimates of human mobility, Journal of The Royal Society Interface 10 (81) (2013) 20120986. doi:10.1098/rsif.2012.0986.

[7] H. Yang, J. Zhou, Optimal traffic counting locations for origin–destination matrix estimation, Transportation Research Part B: Methodological 32 (2) (1998) 109–126. doi:10.1016/s0191-2615(97)00016-7.

[8] E. Castillo, A. J. Conejo, J. M. Menéndez, P. Jiménez, The Observability Problem in Traffic Network Models, Computer-Aided Civil and Infrastructure Engineering 23 (3) (2008) 208–222. doi:10.1111/j.1467-8667.2008.00531.x.

[9] M. Gentili, P. Mirchandani, Locating sensors on traffic networks: Models, challenges and research opportunities, Transportation Research Part C: Emerging Technologies 24 (2012) 227–255. doi:10.1016/j.trc.2012.01.004.

[10] E. Castillo, Z. Grande, A. Calviño, W. Y. Szeto, H. K. Lo, A State-of-the-Art Review of the Sensor Location, Flow Observability, Estimation, and Prediction Problems in Traffic Networks, Journal of Sensors 2015 (2015) 1–26. doi:10.1155/2015/903563.

[11] J. C. Herrera, D. B. Work, R. Herring, X. J. Ban, Q. Jacobson, A. M. Bayen, Evaluation of traffic data obtained via GPS-enabled mobile phones: The Mobile Century field experiment, Transportation Research Part C: Emerging Technologies 18 (4) (2010) 568–583. doi:10.1016/j.trc.2009.10.006.

[12] E. Jenelius, H. N. Koutsopoulos, Travel time estimation for urban road networks using low frequency probe vehicle data, Transportation Research Part B: Methodological 53 (2013) 64–81. doi:10.1016/j.trb.2013.03.008.

[13] A. D. Patire, M. Wright, B. Prodhomme, A. M. Bayen, How much GPS data do we

need?, Transportation Research Part C: Emerging Technologies 58 (2015) 325–342. doi:10.1016/j.trc.2015.02.011.

[14] N.-E. E. Faouzi, H. Leung, A. Kurian, Data fusion in intelligent transportation systems: Progress and challenges – A survey, Information Fusion 12 (1) (2011) 4–10. doi:10.1016/j.inffus.2010.06.001.

[15] Y. Zheng, L. Capra, O. Wolfson, H. Yang, Urban Computing, ACM Transactions on Intelligent Systems and Technology 5 (3) (2014) 1–55. doi:10.1145/2629592.

[16] C. Chen, J. Ma, Y. Susilo, Y. Liu, M. Wang, The promises of big data and small data for travel behavior (aka human mobility) analysis, Transportation Research Part C: Emerging Technologies 68 (2016) 285–299. doi:10.1016/j.trc.2016.04.005.

[17] C. Anda, A. Erath, P. J. Fourie, Transport modelling in the age of big data, International Journal of Urban Sciences 21 (sup1) (2017) 19–42. doi:10.1080/12265934.2017.1281150.

[18] A. Bazzani, B. Giorgini, S. Rambaldi, R. Gallotti, L. Giovannini, Statistical laws in urban mobility from microscopic GPS data in the area of Florence, Journal of Statistical Mechanics: Theory and Experiment 2010 (05) (2010) P05001. doi:10.1088/1742-5468/2010/05/p05001.

[19] R. Gallotti, A. Bazzani, S. Rambaldi, M. Barthelemy, A stochastic model of randomly accelerated walkers for human mobility, Nature Communications 7 (1) (2016) 12600. doi:10.1038/ncomms12600.

[20] C. Mizzi, A. Fabbri, S. Rambaldi, F. Bertini, N. Curti, S. Sinigardi, R. Luzi, G. Venturi, M. Davide, G. Muratore, A. Vannelli, A. Bazzani, Unraveling pedestrian mobility on a road network using ICTs data during great tourist events, EPJ Data Science 7 (1) (2018) 44. doi:10.1140/epjds/s13688-018-0168-2.

[21] C. Mizzi, A. Fabbri, G. Colombini, F. Bertini, A. Bazzani, A survival model to explain the statistical properties of multimodal mobility, Journal of Statistical Mechanics: Theory and Experiment 2022 (2) (2022) 023404. doi:10.1088/1742-5468/ac4c40.

[22] C. Mizzi, A. Baroncini, A. Fabbri, D. Micheli, A. Vannelli, C. Criminisi, S. Jean, A. Bazzani, Individual mobility deep insight using mobile phones data, EPJ Data Science 12 (1) (2023) 56. doi:10.1140/epjds/s13688-023-00431-4.

[23] F. Bellisardi, Data analytics and predictive models for sustainable multimodal mobility in future smart cities, Ph.D. thesis, Alma Mater Studiorum Università di Bologna (2025). doi:10.48676/unibo/amsdottorato/12316.

[24] M. S. Iqbal, C. F. Choudhury, P. Wang, M. C. González, Development of origin–destination matrices using mobile phone call data, Transportation Research Part C: Emerging Technologies 40 (2014) 63–74. doi:10.1016/j.trc.2014.01.002.

[25] L. Alexander, S. Jiang, M. Murga, M. C. González, Origin–destination trips by purpose and time of day inferred from mobile phone data, Transportation Research Part C: Emerging Technologies 58 (2015) 240–250. doi:10.1016/j.trc.2015.02.018.

[26] J. G. WARDROP, ROAD PAPER. SOME THEORETICAL ASPECTS OF ROAD TRAFFIC RESEARCH., Proceedings of the Institution of Civil Engineers 1 (3) (1952) 325–362. doi:10.1680/ipeds.1952.11259.

[27] P. Wang, T. Hunter, A. M. Bayen, K. Schechtner, M. C. González, Understanding Road Usage Patterns in Urban Areas, Scientific Reports 2 (1) (2012) 1001. doi:10.1038/srep01001.

[28] E. Cascetta, Estimation of trip matrices from traffic counts and survey data: A generalized least squares estimator, Transportation Research Part B: Methodological 18 (4-5) (1984) 289–299. doi:10.1016/0191-2615(84)90012-2.

[29] H. Yang, T. Sasaki, Y. Iida, Y. Asakura, Estimation of origin-destination matrices from link traffic counts on congested networks, Transportation Research Part B: Methodological 26 (6) (1992) 417–434. doi:10.1016/0191-2615(92)90008-k.

[30] C. Antoniou, J. Barceló, M. Breen, M. Bullejos, J. Casas, E. Cipriani, B. Ciuffo, T. Djukic, S. Hoogendoorn, V. Marzano, L. Montero, M. Nigro, J. Perarnau, V. Punzo, T. Toledo, H. van Lint, Towards a generic benchmarking platform for origin–destination flows

estimation/updating algorithms: Design, demonstration and validation, Transportation Research Part C: Emerging Technologies 66 (2016) 79–98. doi:10.1016/j.trc.2015.08.009.

[31] G. Flötteröd, M. Bierlaire, K. Nagel, Bayesian Demand Calibration for Dynamic Traffic Simulations, Transportation Science 45 (4) (2011) 541–561. doi:10.1287/trsc.1100.0367.

[32] Y. Wang, M. Papageorgiou, Real-time freeway traffic state estimation based on extended Kalman filter: a general approach, Transportation Research Part B: Methodological 39 (2) (2005) 141–167. doi:10.1016/j.trb.2004.03.003.

[33] D. B. Work, S. Blandin, O. P. Tossavainen, B. Piccoli, A. M. Bayen, A Traffic Model for Velocity Data Assimilation, Applied Mathematics Research eXpress (2010). doi:10.1093/amrx/abq002.

[34] C. Bachmann, B. Abdulhai, M. J. Roorda, B. Moshiri, A comparative assessment of multi-sensor data fusion techniques for freeway traffic speed estimation using microsimulation modeling, Transportation Research Part C: Emerging Technologies 26 (2013) 33–48. doi:10.1016/j.trc.2012.07.003.

[35] T. Seo, T. Kusakabe, Probe vehicle-based traffic state estimation method with spacing information and conservation law, Transportation Research Part C: Emerging Technologies 59 (2015) 391–403. doi:10.1016/j.trc.2015.05.019.

[36] A. Nantes, D. Ngoduy, A. Bhaskar, M. Miska, E. Chung, Real-time traffic state estimation in urban corridors from heterogeneous data, Transportation Research Part C: Emerging Technologies 66 (2016) 99–118. doi:10.1016/j.trc.2015.07.005.

[37] T. Seo, A. M. Bayen, T. Kusakabe, Y. Asakura, Traffic state estimation on highway: A comprehensive survey, Annual Reviews in Control 43 (2017) 128–151. doi:10.1016/j.arcontrol.2017.03.005.

[38] C. Lin, S.-L. Tung, H.-T. Su, W. H. Hsu, Tel2Veh: Fusion of Telecom Data and Vehicle Flow to Predict Camera-Free Traffic via a Spatio-Temporal Framework, in:

Companion Proceedings of the ACM Web Conference 2024, ACM, 2024, pp. 1083–1086. doi:10.1145/3589335.3651442.

[39] D. Lei, M. Xu, S. Wang, A conditional diffusion model for probabilistic estimation of traffic states at sensor-free locations, Transportation Research Part C: Emerging Technologies 166 (2024) 104798. doi:10.1016/j.trc.2024.104798.

[40] N. Maiti, M. Seppecher, L. Leclercq, Scaling methods for estimating macroscopic fundamental diagrams in urban networks with sparse stationary sensor coverage, Transportation Research Part C: Emerging Technologies 178 (2025) 105213. doi:10.1016/j.trc.2025.105213.

[41] A. Bassolas, J. J. Ramasco, R. Herranz, O. G. Cantú-Ros, Mobile phone records to feed activity-based travel demand models: MATSim for studying a cordon toll policy in Barcelona, Transportation Research Part A: Policy and Practice 121 (2019) 56–74. doi:10.1016/j.tra.2018.12.024.

[42] J. Li, E. Rombaut, L. Vanhaverbeke, Agent-based digital traffic model generation for regions facing data scarcity using aggregated cellphone data: a case study for Brussels, International Journal of Digital Earth 17 (1) (2024) 2407046. doi:10.1080/17538947.2024.2407046.

[43] K. Bochenina, S. Agriesti, C. Roncoli, L. Ruotsalainen, From Urban Data to City-Scale Models: A Review of Traffic Simulation Case Studies, IET Intelligent Transport Systems 19 (1) (2025) e70021. doi:10.1049/itr2.70021.

[44] J. P. C. Kleijnen, Verification and validation of simulation models, European Journal of Operational Research 82 (1) (1995) 145–162. doi:10.1016/0377-2217(94)00016-6.

[45] R. G. Sargent, Verification and validation of simulation models, Journal of Simulation 7 (1) (2013) 12–24. doi:10.1057/jos.2012.20.

[46] T. Toledo, H. N. Koutsopoulos, Statistical Validation of Traffic Simulation Models, Transportation Research Record: Journal of the Transportation Research Board 1876 (1) (2004) 142–150. doi:10.3141/1876-15.

[47] Y. Hollander, R. Liu, The principles of calibrating traffic microsimulation models, Transportation 35 (3) (2008) 347–362. doi:10.1007/s11116-007-9156-2.

[48] Y. Zhao, K. M. Kockelman, The propagation of uncertainty through travel demand models: An exploratory analysis, The Annals of Regional Science 36 (1) (2002) 145–163. doi:10.1007/s001680200072.

[49] S. Rasouli, H. Timmermans, Uncertainty in travel demand forecasting models: literature review and research agenda, Transportation Letters 4 (1) (2012) 55–73. doi:10.3328/tl.2012.04.01.55-73.

[50] M. Lippi, M. Bertini, P. Frasconi, Short-Term Traffic Flow Forecasting: An Experimental Comparison of Time-Series Analysis and Supervised Learning, IEEE Transactions on Intelligent Transportation Systems 14 (2) (2013) 871–882. doi:10.1109/tits.2013.2247040.

[51] E. I. Vlahogianni, M. G. Karlaftis, J. C. Golias, Short-term traffic forecasting: Where we are and where we're going, Transportation Research Part C: Emerging Technologies 43 (2014) 3–19. doi:10.1016/j.trc.2014.01.005.

[52] OpenStreetMap contributors, OpenStreetMap, road-network source; accessed 27 October 2026 (2026).
URL https://www.openstreetmap.org/

[53] G. Boeing, OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks, Computers, Environment and Urban Systems 65 (2017) 126–139. doi:10.1016/j.compenvurbsys.2017.05.004.

[54] C. Barrington-Leigh, A. Millard-Ball, The world's user-generated road map is more than 80% complete, PLOS ONE 12 (8) (2017) e0180698. doi:10.1371/journal.pone.0180698.

[55] E. W. Dijkstra, A note on two problems in connexion with graphs, Numerische Mathematik 1 (1) (1959) 269–271. doi:10.1007/bf01386390.

[56] A. Horni, K. Nagel, K. W. Axhausen (Eds.), The Multi-Agent Transport Simulation MATSim, Ubiquity Press, 2016. doi:10.5334/baw.

[57] P. A. Lopez, E. Wiessner, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flotterod, R. Hilbrich, L. Lucken, J. Rummel, P. Wagner, Microscopic Traffic Simulation using SUMO, in: 2018 21st International Conference on Intelligent Transportation Systems (ITSC), IEEE, 2018, pp. 2575–2582. doi:10.1109/itsc.2018.8569938.

[58] M. Treiber, A. Kesting, Traffic Flow Dynamics, Springer Berlin Heidelberg, 2013. doi:10.1007/978-3-642-32460-4.

[59] Regione Emilia-Romagna, Sistema MTS – flussi di traffico al quarto d'ora, Open-data catalogue minERva, dataset DS-1944, CC BY.
URL https://datacatalog.regione.emilia-romagna.it/catalogCTA/dataset/sistema-mts-flussi-di-traffico-al-quarto-d-ora

[60] F. Bellisardi, multi-sensor-traffic-reconstruction: code and aggregate results, GitHub repository (2026).
URL https://github.com/federicobellisardi/multi-sensor-traffic-reconstruction